ACME's systems illustrate how different contexts demand different trade-offs:

- The Customer Collaboration platform relies primarily on token-based signals, favoring performance over immediacy.
- Employees on that same platform use richer context that combines tokens, device-trust attestations, and cached internal data.
- The engineering deployment tools demand the freshest context—real-time checks for on-call status, device posture, and operational readiness.

Later, in Chapter 15, we'll revisit how an organization's security architecture and zero-trust strategy shape these same design choices. Assurance requirements, risk appetite, and governance policies all influence where each system can safely operate within this three-dimensional space. Table 12.1 summarizes how each of these systems occupies a distinct point in the choice space, balancing accuracy, latency, and complexity.

Table 12.1 ACME's trust signals across different contexts. Each system occupies a different point within the three-dimensional choice space—balancing accuracy, latency, and complexity to align with its risk profile. The Customer Collaboration platform prioritizes performance with token-based context; employees use a hybrid approach combining stable and short-lived signals; and the engineering tools depend on high-assurance, real-time data to avoid stale or unauthorized deployments. The freshness times in the table are typical and may vary depending on actual needs.

Context	Key signals	Delivery method	Freshness model
Customers on the Customer Collaboration platform	Customer ID, role, account status, fraud score	OAuth token (ID + role); customer authn system API query for status	Token = 1 hour; account/fraud data refreshed every few hours
Employees on the Customer Collaboration platform	Employee identity, HR status, department, Finance limit, device posture	Internal API calls (HR, Finance); signed device-trust token from posture client	HR/Finance cached daily; device-trust token expires in 15 minutes
Engineers using deployment tools	Employee identity, on-call status, device posture, change ticket approval, CI/CD build verification	Internal identity service; real-time PagerDuty API; signed device-posture token; CI/CD and change-management APIs	On-call and posture data ≤ 5 minutes; others cached daily